

GameFi Economy Protocol

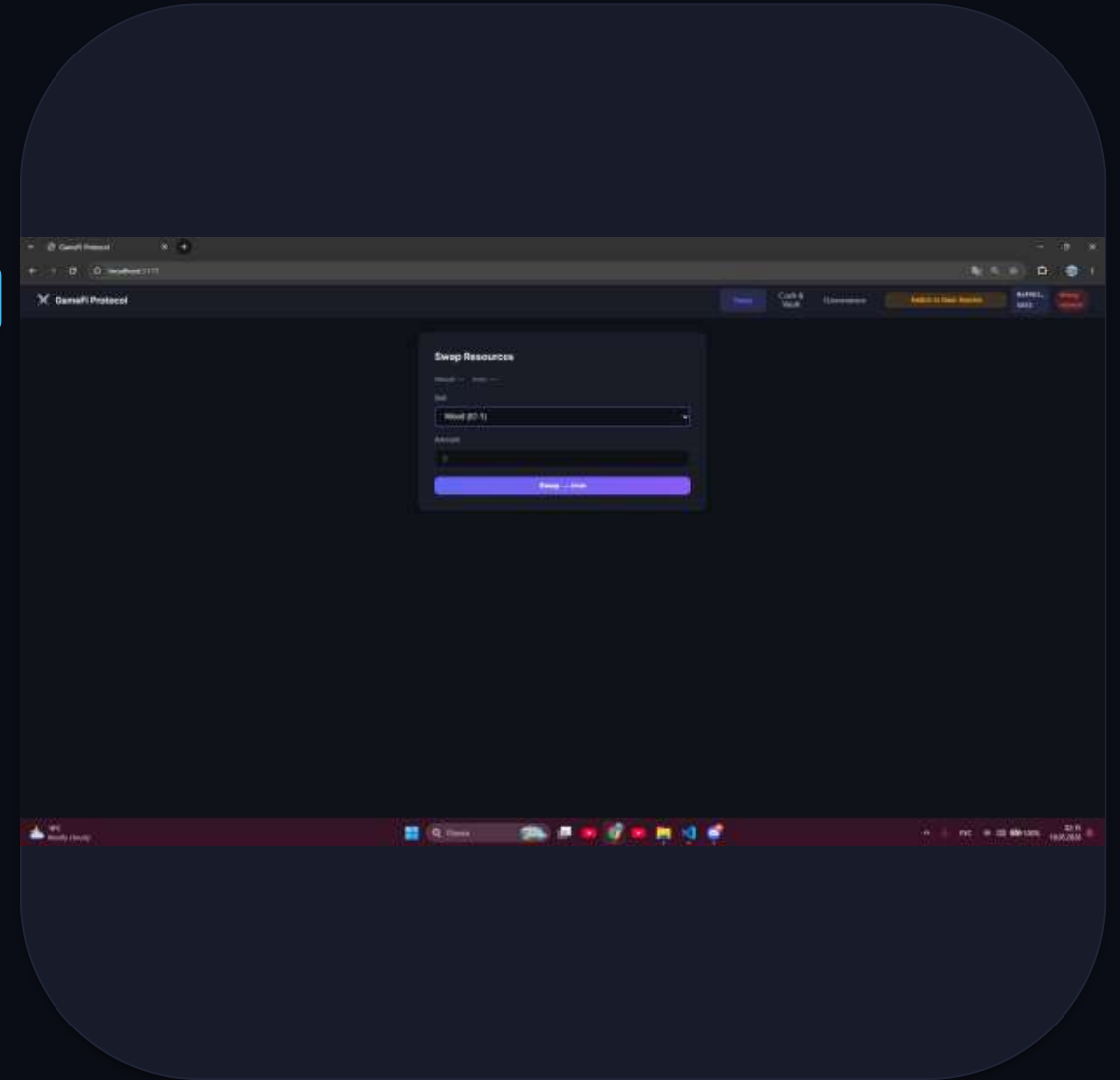
On-chain economy layer for blockchain games

Option B: GameFi
Economy

Arbitrum Sepolia

React + Foundry + The Graph

- ? ERC1155 resources and crafted items
- ? Custom constant-product AMM for resource swaps
- ? UUPS-upgradeable crafting and ERC4626 vault
- ? DAO governance with Governor + Timelock



Problem & Scope

A reusable economy layer, not a full game client

Game economies need

- ? transparent ownership
- ? player-driven markets
- ? secure item lifecycle
- ? upgrade path after launch

Our protocol provides

- ? fungible and non-fungible game assets
- ? AMM trading for resources
- ? crafting progression
- ? governed parameters

Demo focus

- ? live wallet connection
- ? network detection
- ? resource swap
- ? crafting transaction

Architecture

Contract modules and external infrastructure

Frontend

React/Vite dApp
MetaMask +
ethers.js

?

Economy Core

GameItems
ERC1155
ResourceAMM
 $x*y=k$
Crafting UUPS
proxy

?

Token & Vault

GameToken
ERC20Votes/Permi
t
GameVault
ERC4626

?

Governance

GameGovernor
TimelockController
GameTreasury

?

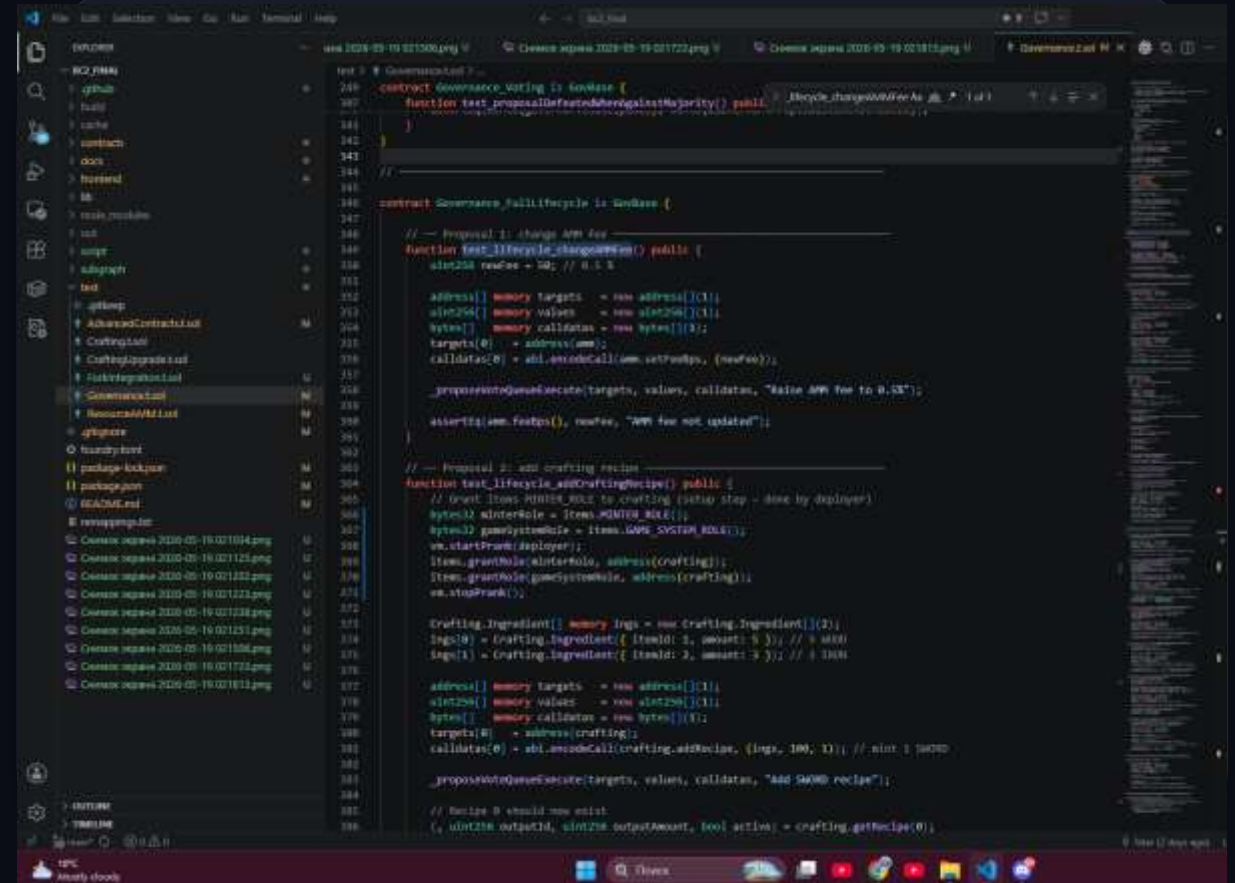
Infra

Chainlink oracle /
VRF flow
The Graph
subgraph
CREATE2 factory

Core Contracts

Standards and responsibilities

- ? GameToken: ERC20Votes + ERC20Permit governance token
- ? GameItems: ERC1155 resources (WOOD, IRON, CRYSTAL) and crafted items
- ? GameVault: ERC4626 deposit/share accounting for GAME
- ? ResourceAMM: custom $x*y=k$ ERC1155 AMM with 0.3% fee and slippage guard
- ? Crafting: UUPS upgradeable recipe system behind ERC1967Proxy



AMM & Crafting Flow

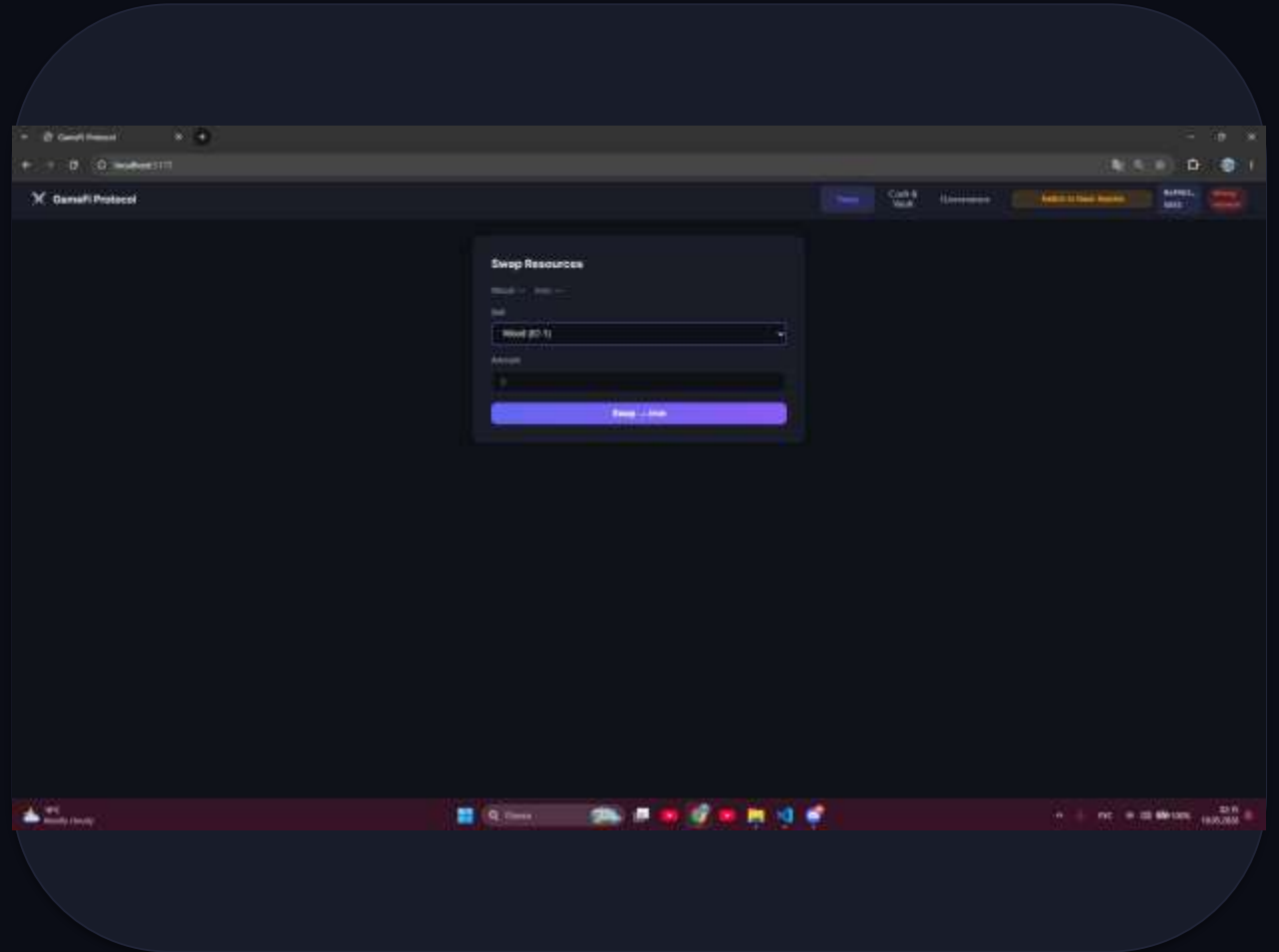
Live demo path

Swap flow

1. User approves ResourceAMM as ERC1155 operator
2. `swap(tokenIn, amountIn, minOut)` executes
3. AMM applies fee and updates reserves
4. User receives opposite resource

Craft flow

Recipe #0: 3 WOOD + 2 IRON -> 1 SWORD



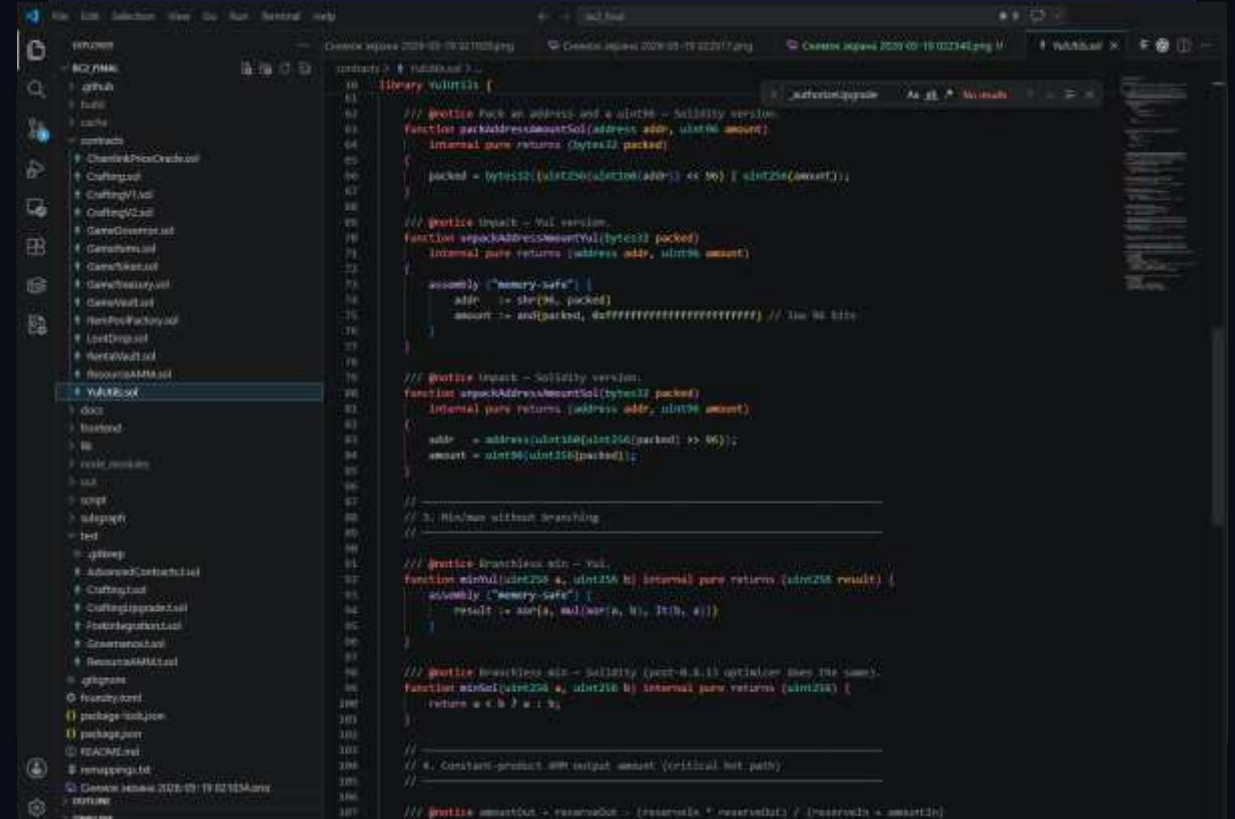
DAO Governance

Governor + TimelockController

- ? Voting token: GAME with ERC20Votes checkpoints
- ? Voting delay: 7,200 blocks, about 1 day
- ? Voting period: 50,400 blocks, about 1 week
- ? Quorum: 4%; proposal threshold: 1%
- ? Timelock delay: 2 days before execution

Propose -> Vote -> Queue -> Execute

Timelock owns treasury actions and delayed parameter changes.



```
10 library Timelock {
11
12     /// @notice Pack an address and a uint96 - Solidity version.
13     function packAddressAmountSol(address addr, uint96 amount)
14         internal pure returns (bytes32 packed) {
15
16         packed = bytes32(uint256(uint(addr)) << 96) | uint256(amount);
17
18     }
19
20     /// @notice Unpack - Solidity version.
21     function unpackAddressAmountSol(bytes32 packed)
22         internal pure returns (address addr, uint96 amount) {
23
24         assembly ("memory-safe") {
25             addr := shr(96, packed)
26             amount := and(packed, 0xffffffffffffffff) // low 96 bits
27         }
28
29     }
30
31     /// @notice Input - Solidity version.
32     function input(bytes32 packed)
33         internal pure returns (address addr, uint96 amount) {
34
35         addr = address(uint160(uint(packed) >> 96));
36         amount = uint96(uint256(packed));
37
38     }
39
40     // -----
41     // 3. Min/max without branching
42     // -----
43
44     /// @notice Branchless min - Solidity (post-0.8.15 optimizer does the same).
45     function minSol(uint256 a, uint256 b) internal pure returns (uint256 result) {
46
47         assembly ("memory-safe") {
48             result := and(a, 0xffffffff) | and(b, 0xffffffff)
49         }
50
51     }
52
53     /// @notice Branchless min - Solidity (post-0.8.15 optimizer does the same).
54     function minSol(uint256 a, uint256 b) internal pure returns (uint256) {
55
56         return a < b ? a : b;
57
58     }
59
60     // -----
61     // 4. Constant-product with output amount (critical hot path)
62     // -----
63
64     /// @notice amountOut -> reserveOut - (reserveIn * reserveOut) / (reserveIn * amountIn)
```

Advanced Requirements

Mandatory course components covered

UUPS proxy

CraftingV1 -> CraftingV2 upgrade path

CREATE /
CREATE2

ItemPoolFactory standard and deterministic pool deployment

Yul assembly

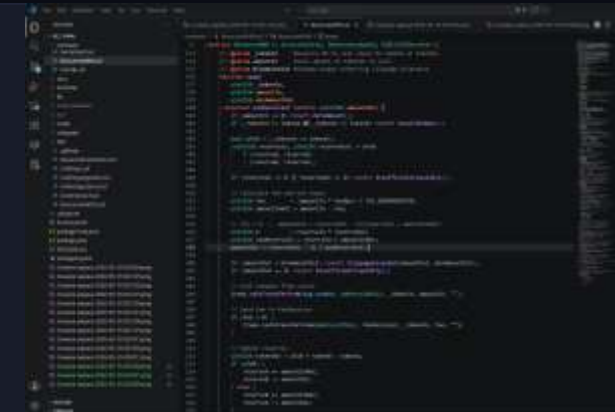
YulUtils and AMM sqrt math

Chainlink

Price feed wrapper with staleness checks; VRF-compatible loot flow

The Graph

Subgraph schema for swaps, crafting, rentals, loot and governance



Testing & Verification

Foundry, frontend, subgraph and security tooling

- ? Unit, fuzz, invariant and optional fork tests
- ? Current local snapshot: 87 passed, 0 failed, 3 skipped
- ? Frontend build passes with React/Vite
- ? Subgraph build passes with generated ABIs and mappings
- ? Slither high-severity gate included in CI



Deployment

Foundry broadcast output and frontend wiring

Network

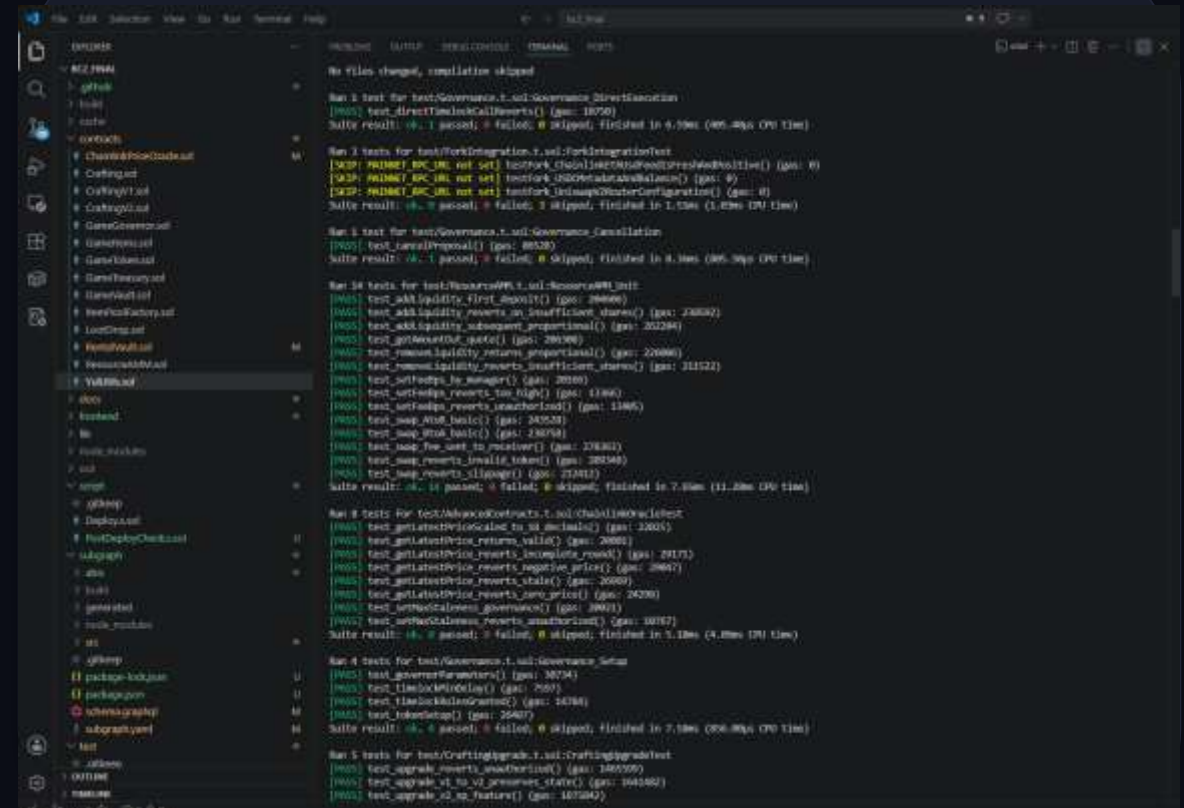
Arbitrum Sepolia
Chain ID: 421614

ResourceAMM: 0x69e951e9cf90c8d116b59276b3b68ae548fa9920
Crafting proxy:
0x9d2b2608cd9cb1abfe026f5800d5f0f6f5cf427b
GameItems: 0x044b55374010b4b06cb39afdc88f8c7444258f87

? Addresses copied from run-latest.json

? Frontend checks chain and prompts switch

? DemoSetup seeded resources, AMM liquidity and recipe
#0



```
Run 1 test for test/Governance.t.sol:Governance:DirectRejection
[PASS] test_directRejectionCallReverts() (gas: 10720)
Suite result: ok, 1 passed, 0 failed, 0 skipped, finished in 4.50ms (405.40µs CPU time)

Run 2 tests for test/ForkIntegration.t.sol:ForkIntegrationTest
[SKIP: MAINNET_RPC_URL not set] testFork_ChainlinkNodeFeedForwardNodePosTime() (gas: 0)
[SKIP: MAINNET_RPC_URL not set] testFork_ChainlinkNodeFeedForwardNodeBalance() (gas: 0)
[SKIP: MAINNET_RPC_URL not set] testFork_ChainlinkNodeFeedForwardNodeConfiguration() (gas: 0)
Suite result: ok, 0 passed, 0 failed, 0 skipped, finished in 1.13ms (1.03ms CPU time)

Run 3 test for test/Governance.t.sol:Governance:Cancelation
[PASS] test_cancelRejection() (gas: 80320)
Suite result: ok, 1 passed, 0 failed, 0 skipped, finished in 4.1ms (305.30µs CPU time)

Run 34 tests for test/ResourceAMM.t.sol:ResourceAMM:Unit
[PASS] test_addLiquidity_FirstDeposit() (gas: 286000)
[PASS] test_addLiquidity_reverts_on_insufficient_funds() (gas: 23030)
[PASS] test_addLiquidity_subsequent_proportional() (gas: 251294)
[PASS] test_getResourceQuote_quote() (gas: 200300)
[PASS] test_removeLiquidity_return_proportional() (gas: 220000)
[PASS] test_removeLiquidity_reverts_insufficient_stakes() (gas: 211522)
[PASS] test_setFeesByManager() (gas: 20100)
[PASS] test_setFees_reverts_too_high() (gas: 1186)
[PASS] test_setFees_reverts_unauthorized() (gas: 1186)
[PASS] test_swap_strk_basic() (gas: 240520)
[PASS] test_swap_strk_buile() (gas: 240794)
[PASS] test_swap_strk_sell_to_buyer() (gas: 238402)
[PASS] test_swap_reverts_invalid_token() (gas: 201540)
[PASS] test_swap_reverts_slippage() (gas: 272812)
Suite result: ok, 14 passed, 0 failed, 0 skipped, finished in 7.01ms (11.20ms CPU time)

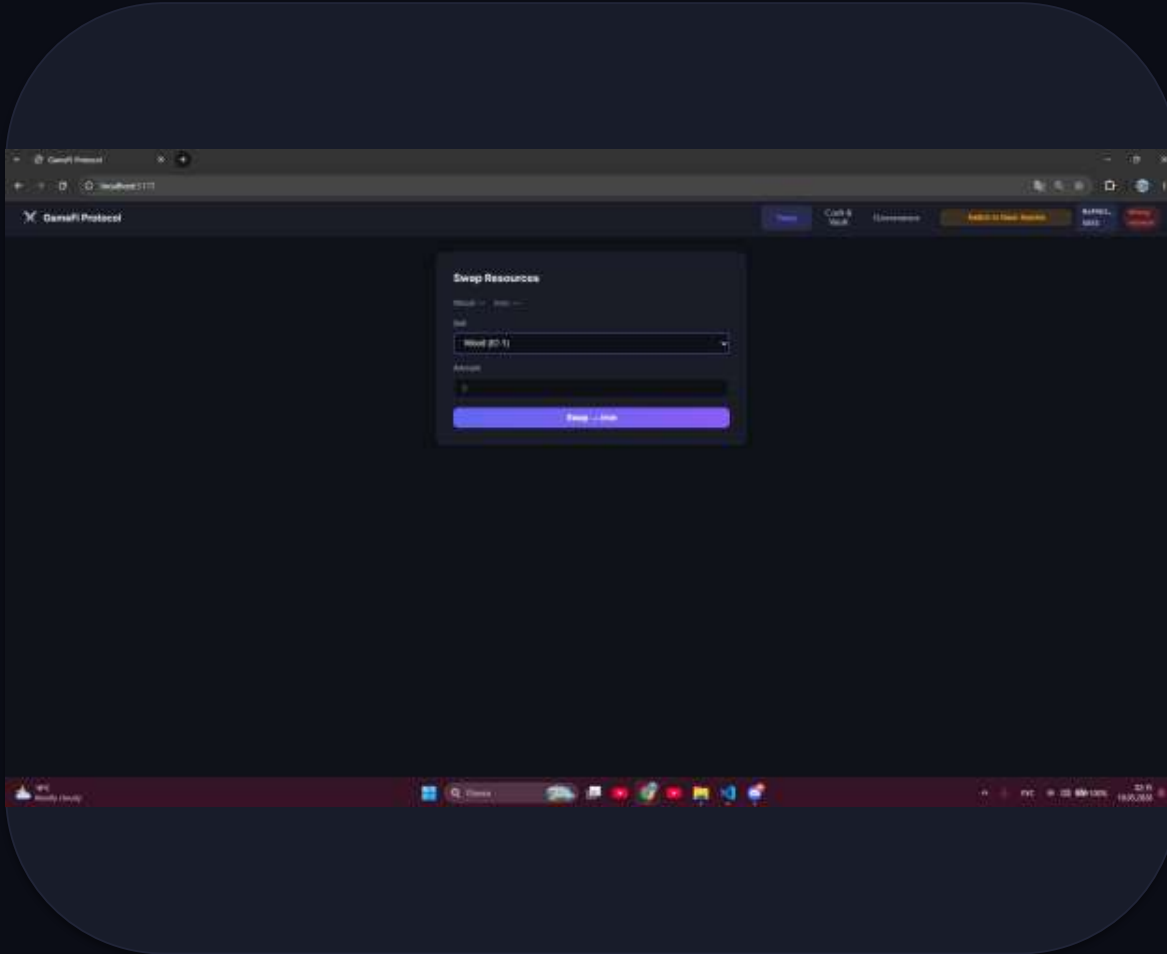
Run 8 tests for test/AdvancedContracts.t.sol:ChainlinkOracleTest
[PASS] test_generatePriceValidIn30seconds() (gas: 33005)
[PASS] test_generatePrice_return_valid() (gas: 20000)
[PASS] test_generatePrice_reverts_incomplete_round() (gas: 29171)
[PASS] test_generatePrice_reverts_negative_price() (gas: 20847)
[PASS] test_generatePrice_reverts_stale() (gas: 20000)
[PASS] test_generatePrice_reverts_zero_price() (gas: 24200)
[PASS] test_generatePrice_reverts_governance() (gas: 30001)
[PASS] test_generatePrice_reverts_unauthorized() (gas: 30167)
Suite result: ok, 8 passed, 0 failed, 0 skipped, finished in 5.11ms (4.0ms CPU time)

Run 4 tests for test/Governance.t.sol:Governance:Setup
[PASS] test_governanceParameters() (gas: 30734)
[PASS] test_lineOfProtocol() (gas: 7669)
[PASS] test_lineOfBlockchain() (gas: 24384)
[PASS] test_tokenSale() (gas: 26407)
Suite result: ok, 4 passed, 0 failed, 0 skipped, finished in 7.1ms (590.00µs CPU time)

Run 5 tests for test/CraftingUpgrade.t.sol:CraftingUpgradeTest
[PASS] test_upgrade_reverts_unauthorized() (gas: 1400300)
[PASS] test_upgrade_v1_to_v2_governance_upgrade() (gas: 104582)
[PASS] test_upgrade_v2_to_features() (gas: 107082)
```

Frontend Demo

MetaMask + deployed contracts



Live demo checklist

1. Connect MetaMask
2. Switch to Arbitrum Sepolia
3. Read WOOD/IRON balances and pool reserves
4. Swap a small amount: 1 or 2
5. Open Crafting and craft recipe #0

Avoid Vault unless the demo wallet has GAME tokens.

Team Presentation Split

3 speakers, clean responsibility boundaries

Speaker 1

Smart Contracts / Architecture

- ? GameToken, GameItems, GameVault
- ? ResourceAMM and Crafting UUPS
- ? Explain protocol modules

Speaker 2

Governance / Advanced / Tests

- ? Governor + Timelock
- ? CREATE2, Yul, Chainlink, subgraph
- ? Test and security snapshot

Speaker 3

Frontend / Live Demo

- ? MetaMask and network detection
- ? contracts.ts deployment addresses
- ? Swap + Craft live flow

Summary

What the live demo proves

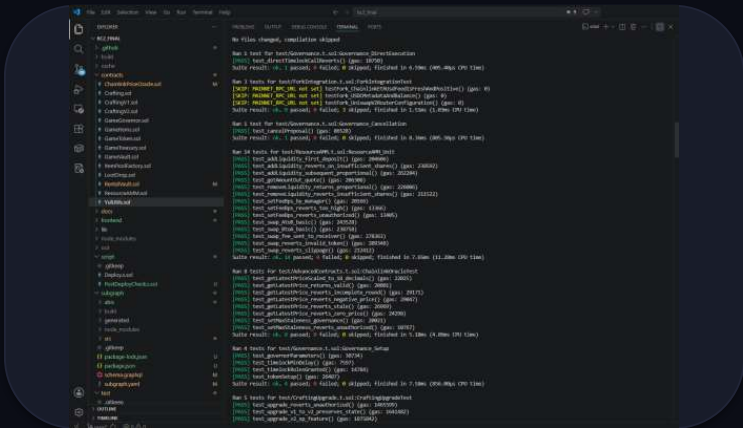
- ? Contracts are deployed on an L2 testnet and wired into the frontend
- ? The dApp reads real ERC1155 balances and AMM reserves
- ? Swap and crafting execute as wallet-confirmed transactions
- ? Architecture covers required standards, governance, oracles, indexing and advanced Solidity
- ? Remaining production work: verification links, real VRF subscription, stronger vault/rental coverage

Q&A

Be ready to explain each module at architecture level.

Appendix: Screenshots

Code, tests, UI and repository evidence



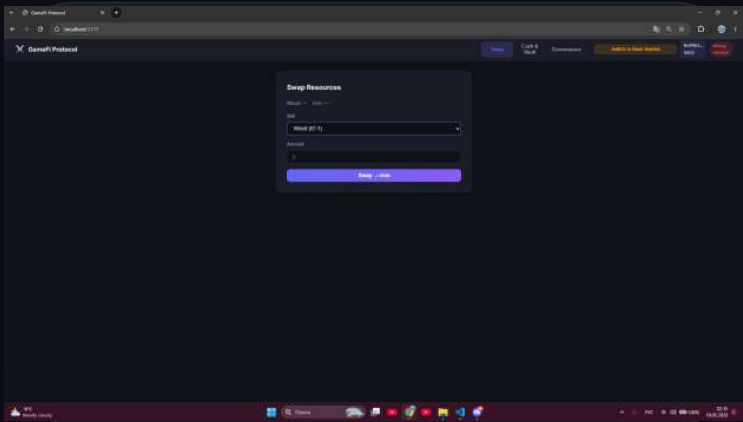
Screenshot 1



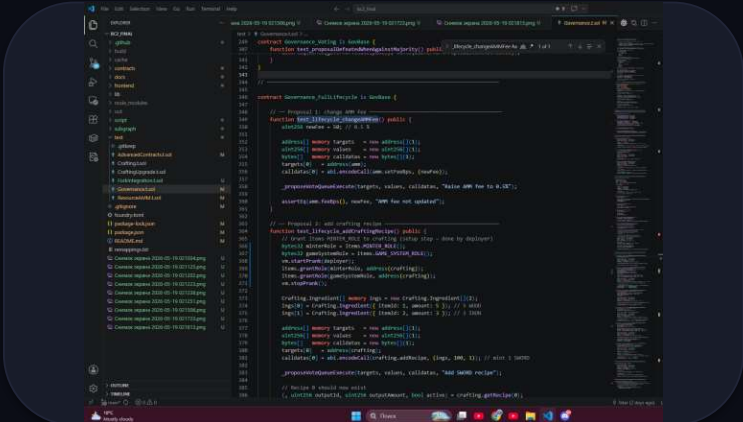
Screenshot 3



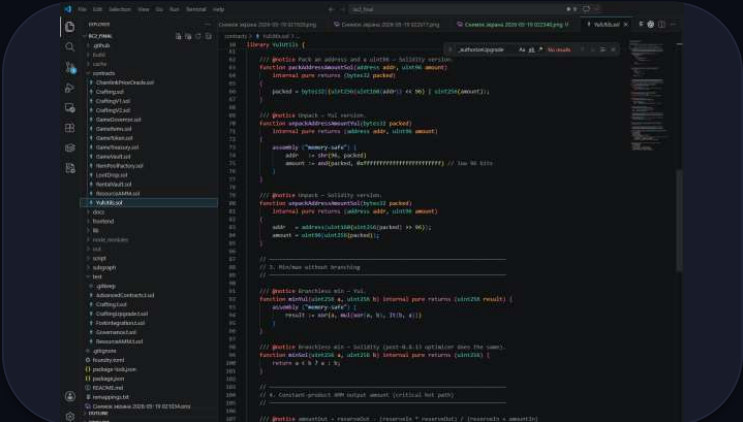
Screenshot 6



Screenshot 7



Screenshot 10



Screenshot 13